

More than genes

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PSYC1003

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Interactions between genetic background and the environment

1. Explain why behaviour cannot usually be attributed to genes or environments alone.
2. Describe how environmental experiences can produce lasting biological changes.
3. Use cross-fostering evidence to distinguish genetic inheritance from developmental experience.
4. Explain why physiological activity does not determine a single psychological experience.
5. Apply Sapolsky's timeline to explain a behaviour at several levels.
6. Critically evaluate biologically deterministic explanations of behaviour.

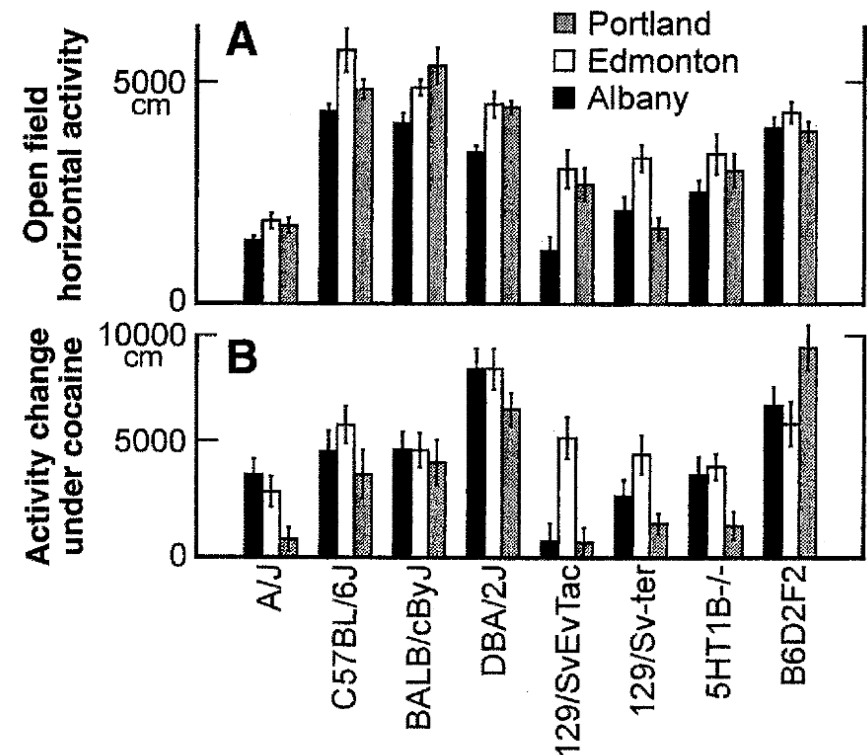
Non-trivial interactions between mouse strains and environments on simple behaviours

Genetics of Mouse Behavior: Interactions with Laboratory Environment

John C. Crabbe,^{1*} Douglas Wahlsten,² Bruce C. Dudek³

Strains of mice that show characteristic patterns of behavior are critical for research in neurobehavioral genetics. Possible confounding influences of the laboratory environment were studied in several inbred strains and one null mutant by simultaneous testing in three laboratories on a battery of six behaviors. Apparatus, test protocols, and many environmental variables were rigorously equated. Strains differed markedly in all behaviors, and despite standardization, there were systematic differences in behavior across labs. For some tests, the magnitude of genetic differences depended upon the specific testing lab. Thus, experiments characterizing mutants may yield results that are idiosyncratic to a particular laboratory.

4 JUNE 1999 VOL 284 SCIENCE www.sciencemag.org



Environmental effects modified by genetic background

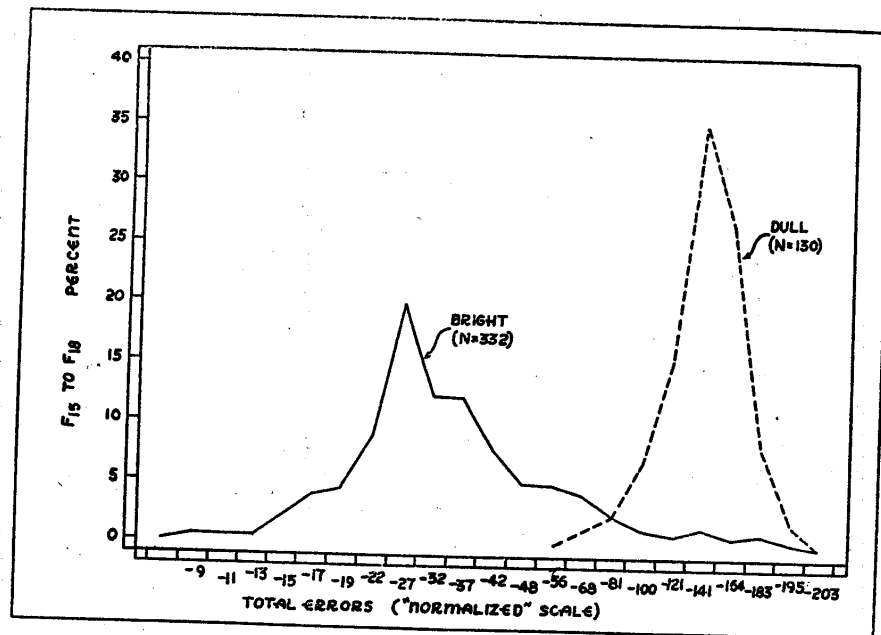
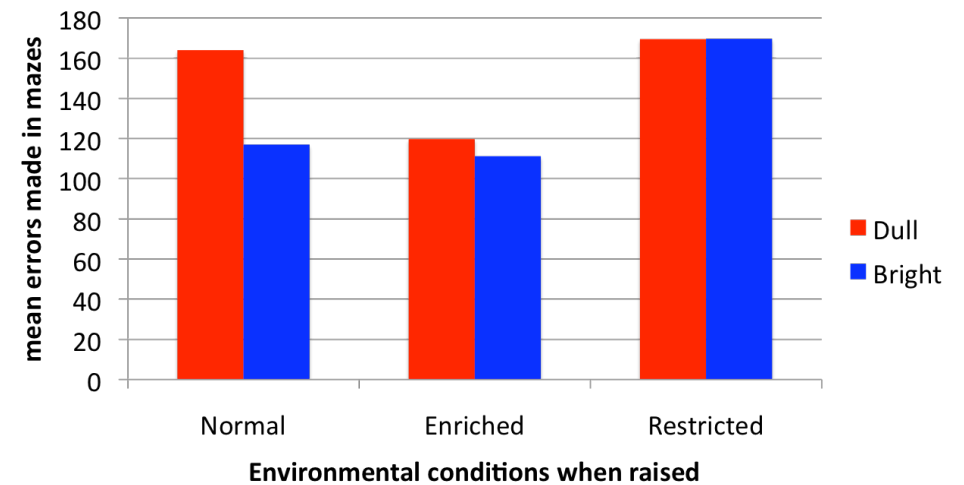


FIG. V.—BRIGHT AND DULL STRAINS OF THE FIFTEENTH TO EIGHTEENTH GENERATIONS



Cooper & Zubek (1958) Can J Psychol, 12: 159-164

Gene activity is modified by the environment - epigenetics

This is a locust

This is a grasshopper



They are the same species, *Schistocerca gregaria*, and are genetically identical.

- Locusts develop when there is a high population density and low food availability, they swarm and become migratory.
- They can change phenotypes rapidly – within hours
- This is accompanied by widespread changes in expression of genes in CNS

Nongenomic Transmission Across Generations of Maternal Behavior and Stress Responses in the Rat

Darlene Francis, Josie Diorio, Dong Liu, Michael J. Meaney*

In the rat, variations in maternal care appear to influence the development of behavioral and endocrine responses to stress in the offspring. The results of cross-fostering studies reported here provide evidence for (i) a causal relationship between maternal behavior and stress reactivity in the offspring and (ii) the transmission of such individual differences in maternal behavior from one generation of females to the next. Moreover, an environmental manipulation imposed during early development that alters maternal behavior can then affect the pattern of transmission in subsequent generations. Taken together, these findings indicate that variations in maternal care can serve as the basis for a nongenomic behavioral transmission of individual differences in stress reactivity across generations.



These two photographs of rodent dams and their pups depict two maternal behaviors that represent a high degree of nurturing for newborn rat pups: (A) “arched back nursing” and (B) “licking and grooming.” Adult offspring of dams that exhibit these behaviors are less fearful than their counterparts reared by less nurturing dams.

(A) **Arched-back nursing.** The dam nursing her pups in this photograph has her back and ventral surface arched, exposing all of her nipples to all pups simultaneously. This makes it easier for each pup find a nipple and remain attached. In contrast, the nipples of a dam lying on her side are relatively less accessible.



(B) **Licking and grooming.** The lower portion of this photograph shows the pink, hairless heads and bodies of several rat pups. The dam, whose fur is black, is licking their skin with great vigor and frequency, constituting a very strong tactile stimulus for these newborn pups. This tactile stimulation promotes of a series of biochemical events that culminate in less fearful adult offspring.

Adoption study in rats

High licking/grooming mums (H) vs Low licking/grooming mums (L)

Adopt out all the offspring of H and L mums at birth

Give half of each group to H or L Mums

Gives 4 groups

H-H, H-L, L-H, L-L

And undisturbed (C) or disturbed (w) controls

H/C, L/C, H/w, L/w

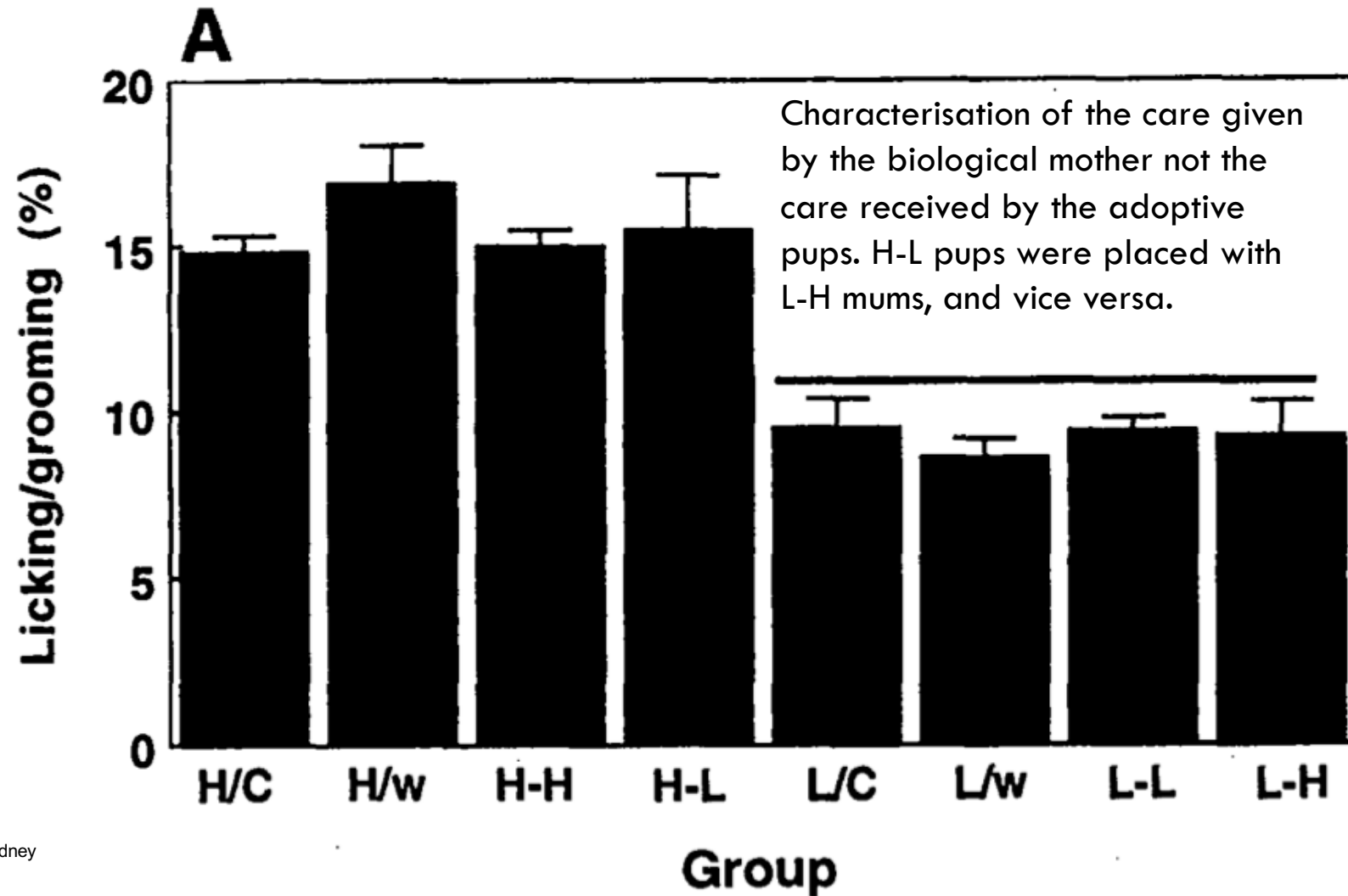
Then examine three things:

What is the rate of licking or grooming of the L or H mums to their adopted or natural offspring?

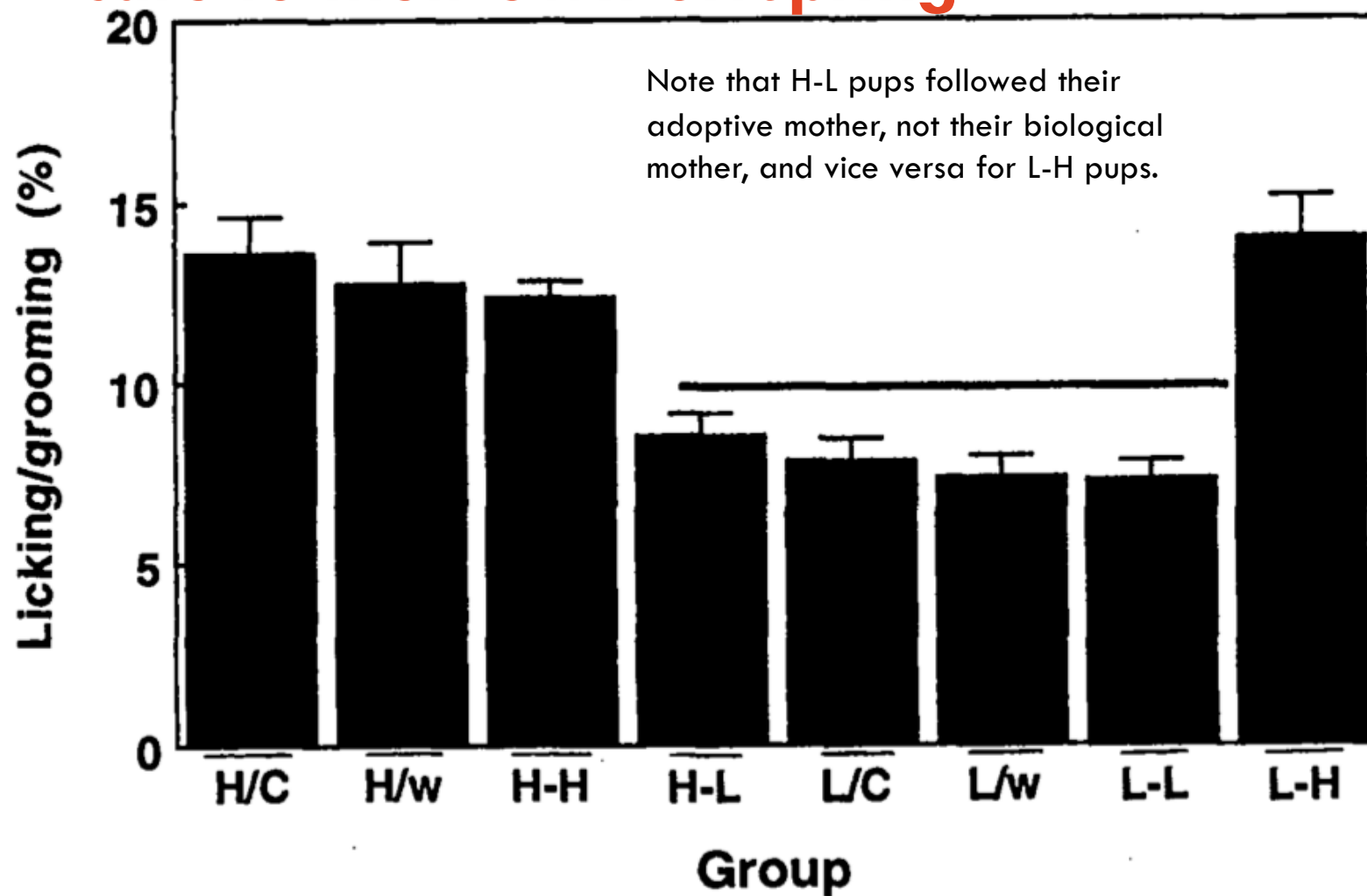
Do the offspring show changes in tests of anxiety as adults?

Do the offspring change the type of maternal care they show their own offspring when they breed?

Rats raised by H mums get better maternal care



Rats raised by H mums give better care to their own offspring



What is the mechanism?

Note, behaviour follows the adopted mother, not the biological mother

- Environment not genes

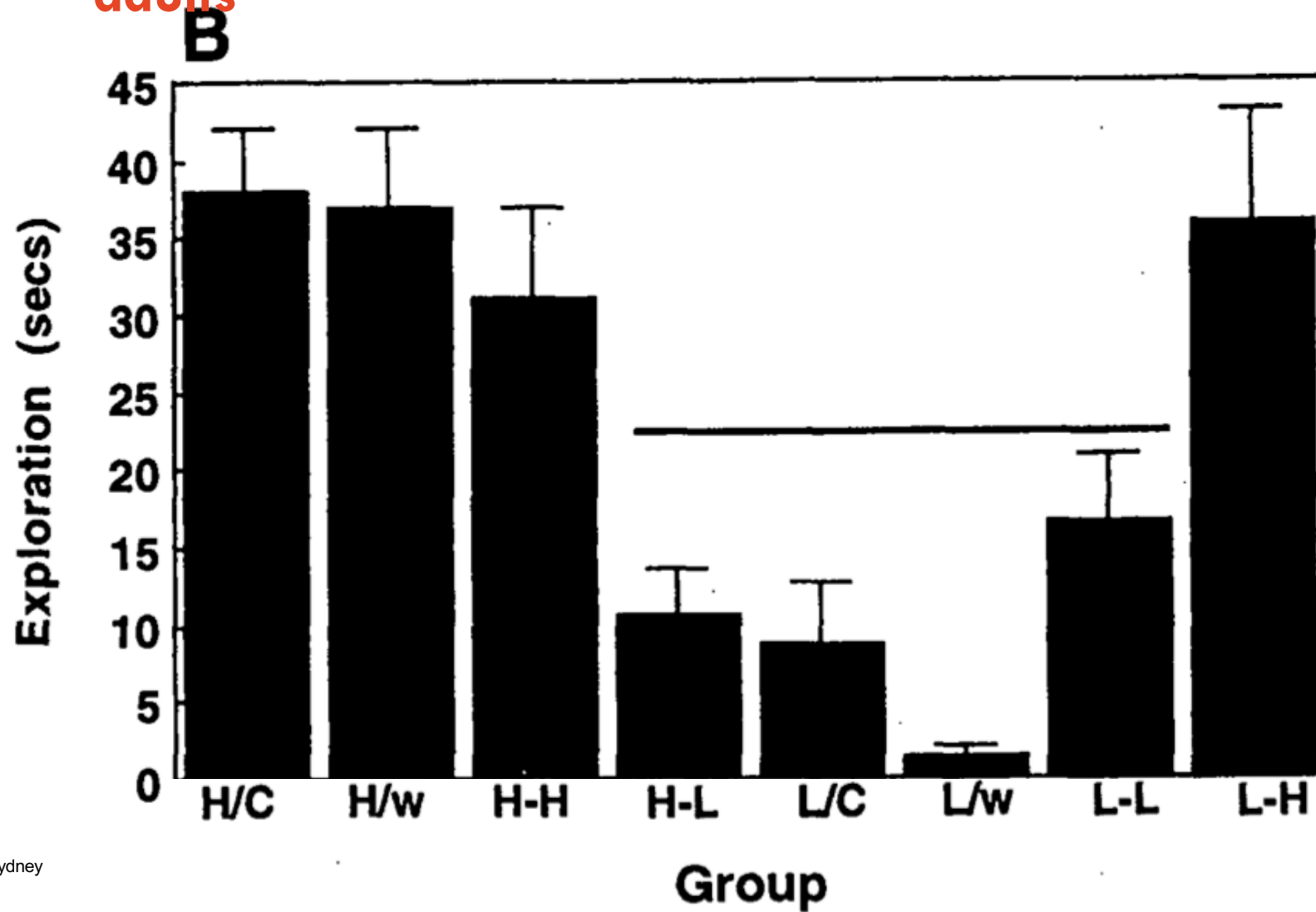
Is it modeling?

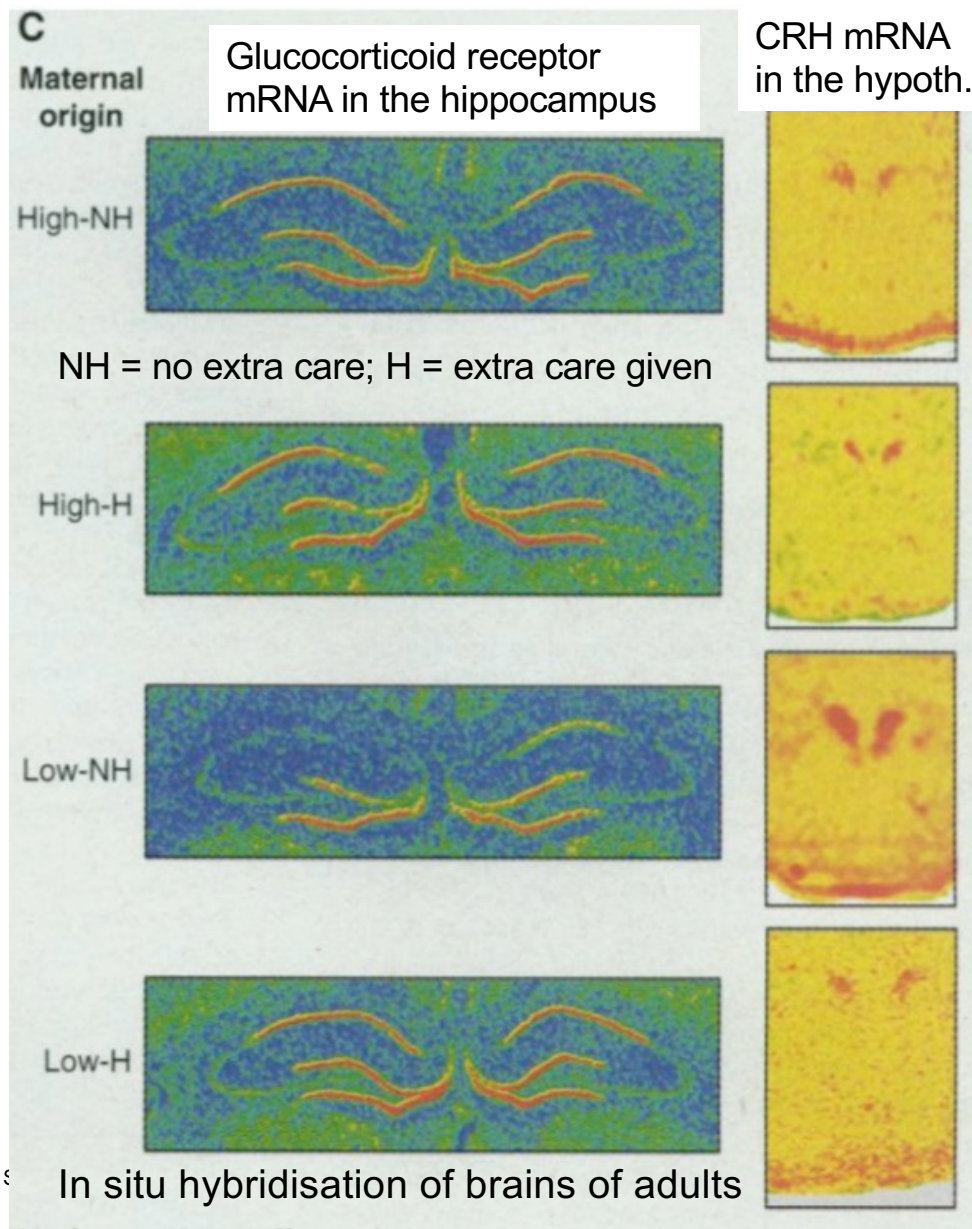
- E.g., Observational learning of maternal behaviours?

Or might the environment be having a more subtle effect?

- Let's look at emotional responses

Rats raised by H mums are less anxious when adults

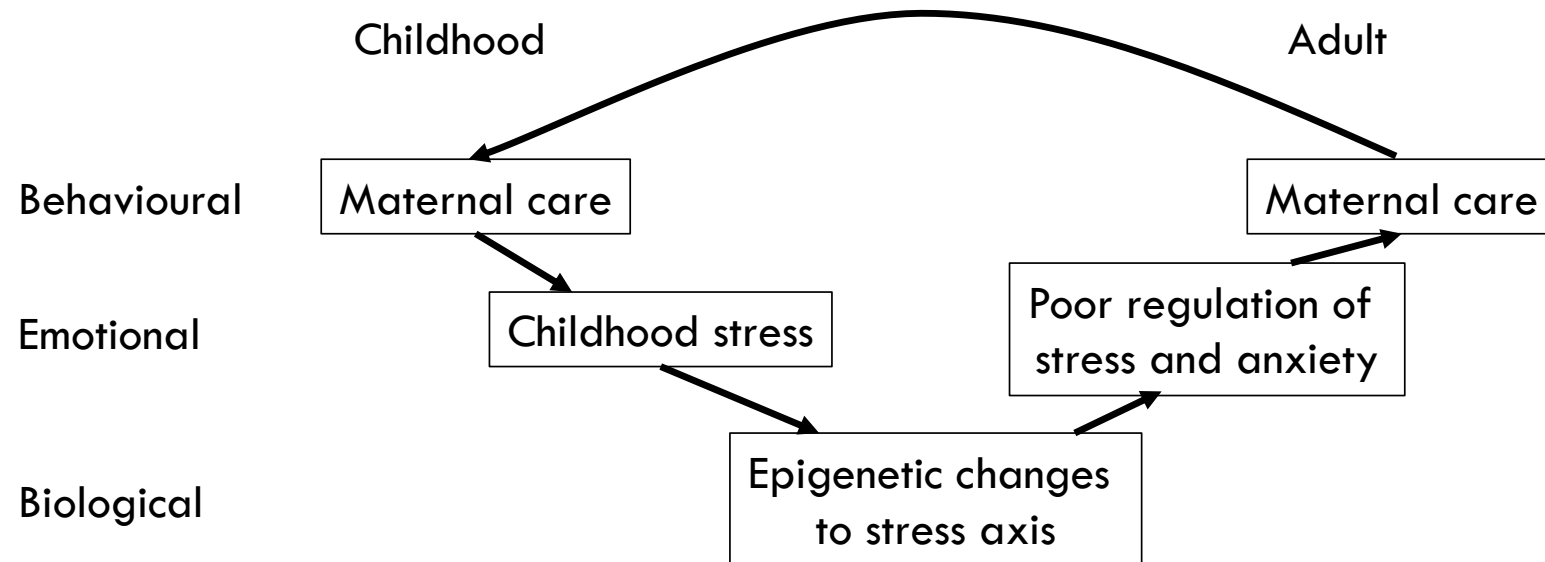




Type of maternal care (and not the genes passed from the biological mother) affects the pattern of gene expression (mRNA) in the stress hormone system in adults.

Early life experience permanently alters genetic activity, regardless of the genotype of the individuals.

Vicious cycle of impoverished care



Note: Not epigenetic regulation of “genes for maternal behaviour”. It’s an indirect effect via emotional regulation.

Subtle and indirect interactions between environment and genes on behaviour

The thing to note here is the indirect effect on maternal behaviour:

Not an environmental effect on “Genes for maternal behaviour”

Effect is on genes important for emotional regulation

Poor emotional regulation then spreads to negatively affect other behaviours, including maternal care

This is one of the reasons why it's so hard to see clear links between genes and behaviours

Variations in a behavioural phenotype might be due to variations in genetic activity of genes important for other factors (e.g., underlying emotional, motivational, cognitive factors)

Variations in these factors might be heavily influenced by environmental context

This is why it's important to understand the psychological factors that contribute to behavioural phenotypes.

PMC full text:

[Transl Psychiatry. 2021; 11: 134.](#)

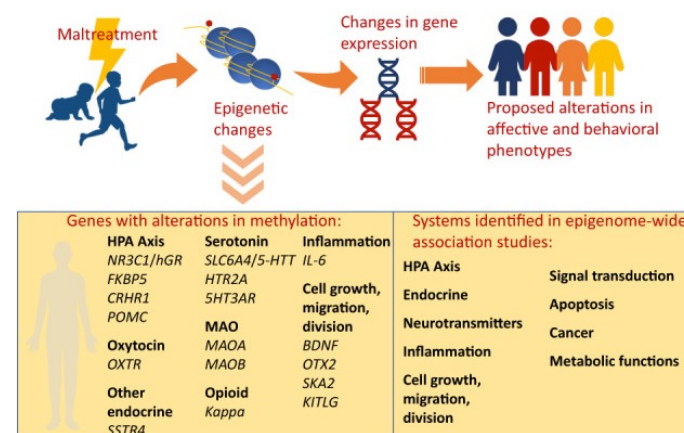
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[<< Prev](#) [Fig. 2](#) [Next >](#)

Fig. 2



Conceptual diagram representing the epigenetic pathway from maltreatment to health and mental health outcomes.

Note: This review presents data focused on the relationship between maltreatment and alterations in methylation, whereas the proposed alterations in affective and behavioral phenotypes are explored further elsewhere.

Biological causes of behaviour

Physiological responses do not determine emotional experiences

- Schacter and Singer, 1962

Subjects told they were getting a drug for an eye test.

- They were actually given adrenaline (or placebo)

Some were told about side effects on heart rate, etc, others were misinformed.

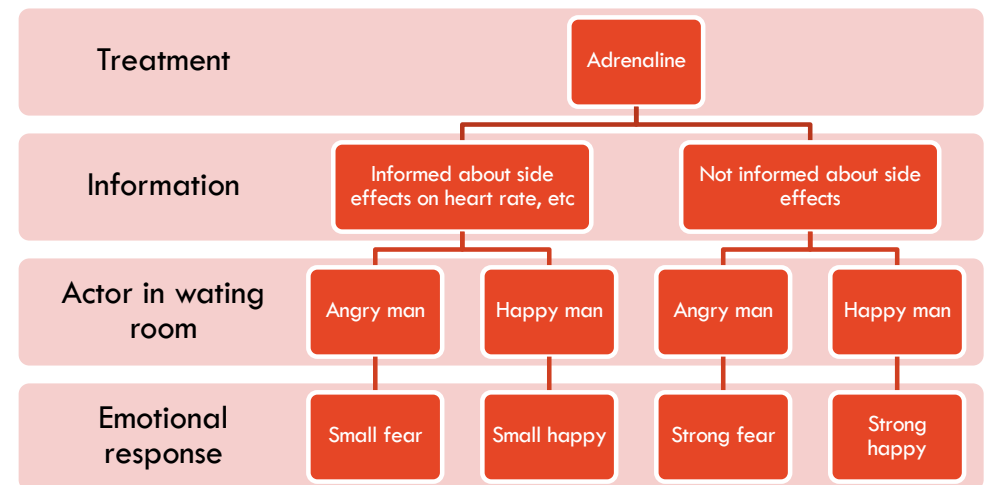
While waiting for the eye test, they saw an actor in the waiting room

- Sometimes he was acting happy
- Other times he was acting angry at waiting

They then did a questionnaire on emotional states.

Misinformed subjects expressed stronger happiness or anger than informed subjects, depending on the context of the waiting room.

- Arousal didn't produce the emotion, the appraisal of the context of arousal did.



Effect of history of stress on emotional responses to arousal

CRF is a hormone released during acute stress

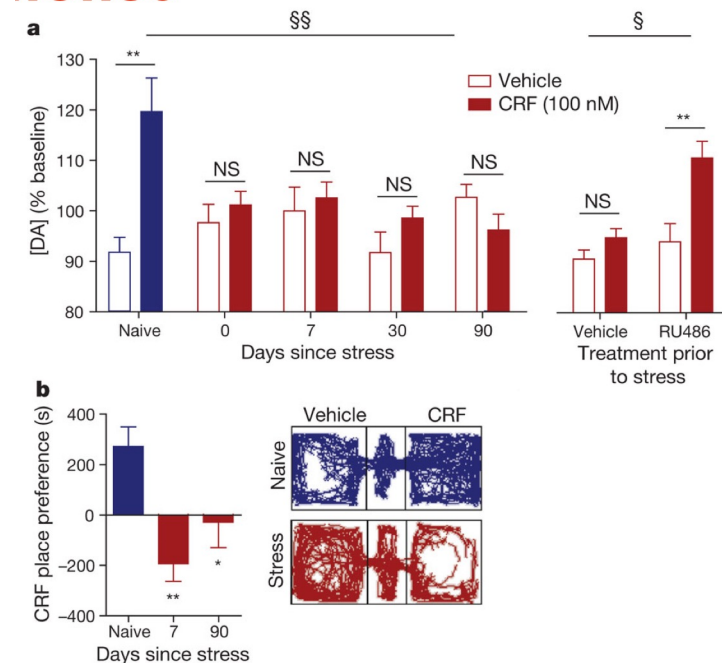
- Important for cortisol response

Injection of mice with CRF typically:

- Releases dopamine (DA) in the brain
- Induces a reward response in a place preference test

However, animals that have a history of exposure to chronic stress show:

- Reduced dopamine response to CRF
- An aversion to places associated with CRF injection



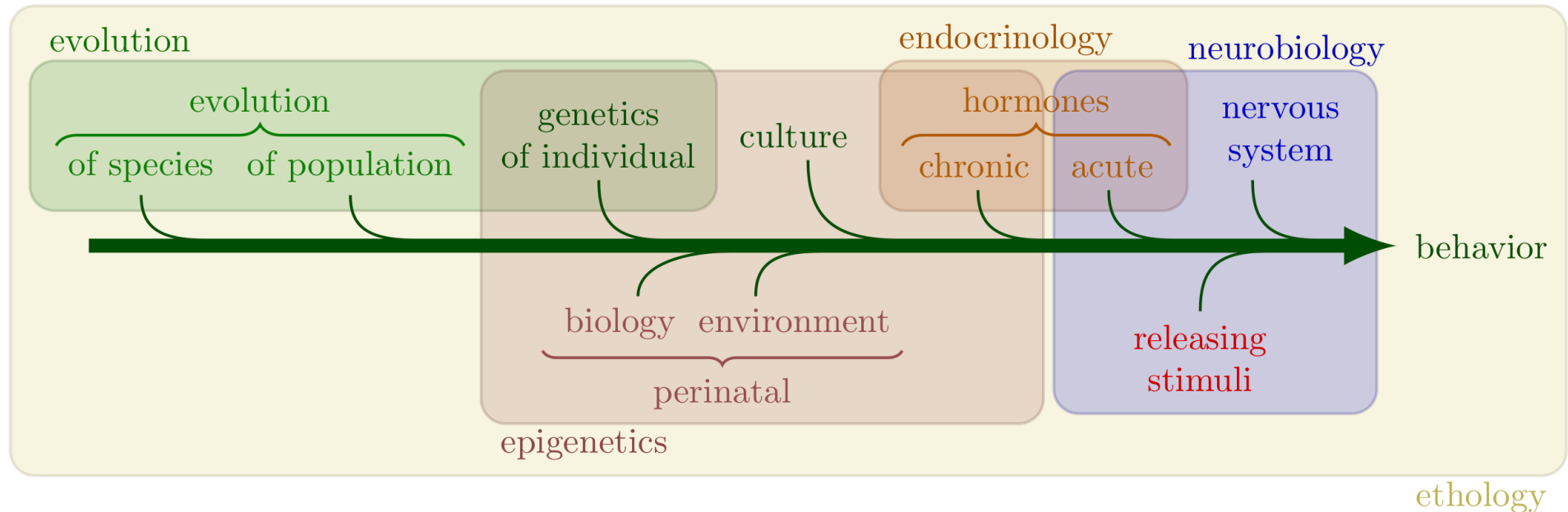
Lemos et al., *Nature* (2012)

The causes of behaviour

Sapolsky (2017):

- No human behaviour has a single cause
 - An action is the endpoint of interacting processes operating over radically different timescales—from neural activity in the preceding second to evolutionary pressures acting over millions of years.
- To explain why someone behaved aggressively, compassionately or courageously, it is therefore inadequate to say:
 - “Their brain caused it.”
 - “It was their hormones.”
 - “It is genetic.”
 - “They were raised that way.”
 - “Humans evolved to behave like that.”
- **Each may describe part of the causal history.**
- **None is sufficient alone.**

Timeline of causes of behaviour

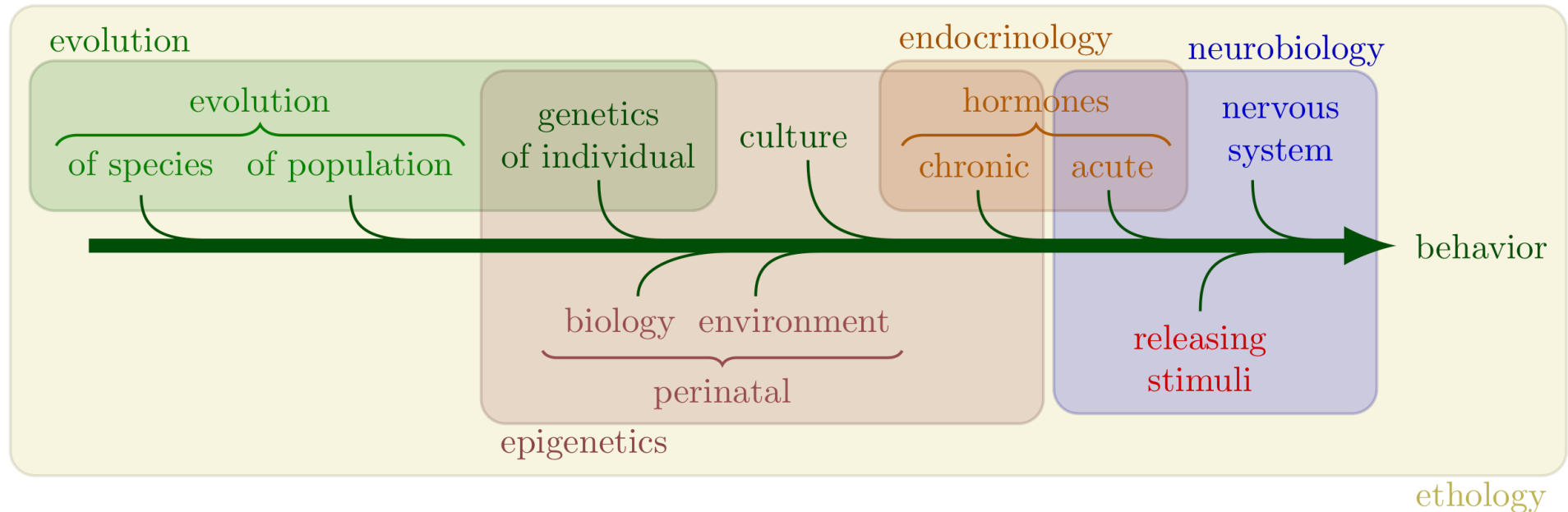


A second before: the nervous system

Immediately before an action, patterns of activity in the brain include:

- Perception and interpretation; emotional and threat responses; memory retrieval; impulse control; anticipation of consequences; selection of an action.

Timeline of causes of behaviour

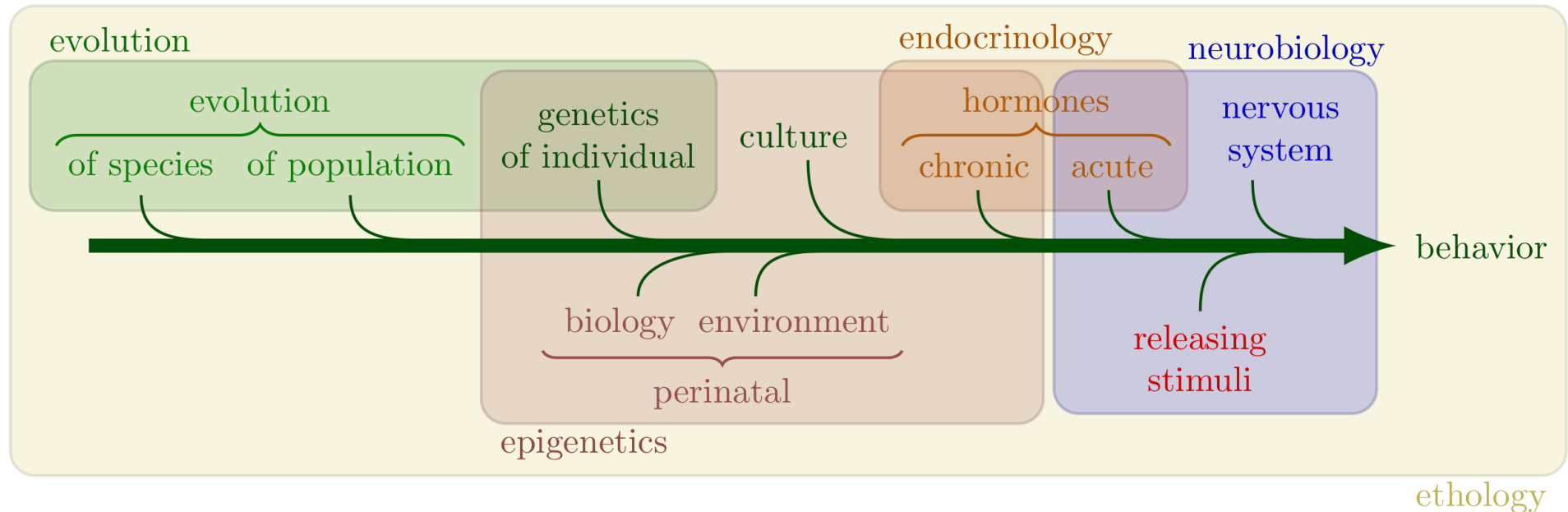


Seconds to minutes before: sensory cues and unconscious priming

The nervous system has just encountered particular sights, sounds, smells, words and social signals. These can:

- Direct attention; activate associations; alter emotional interpretation; make particular responses more accessible.

Timeline of causes of behaviour

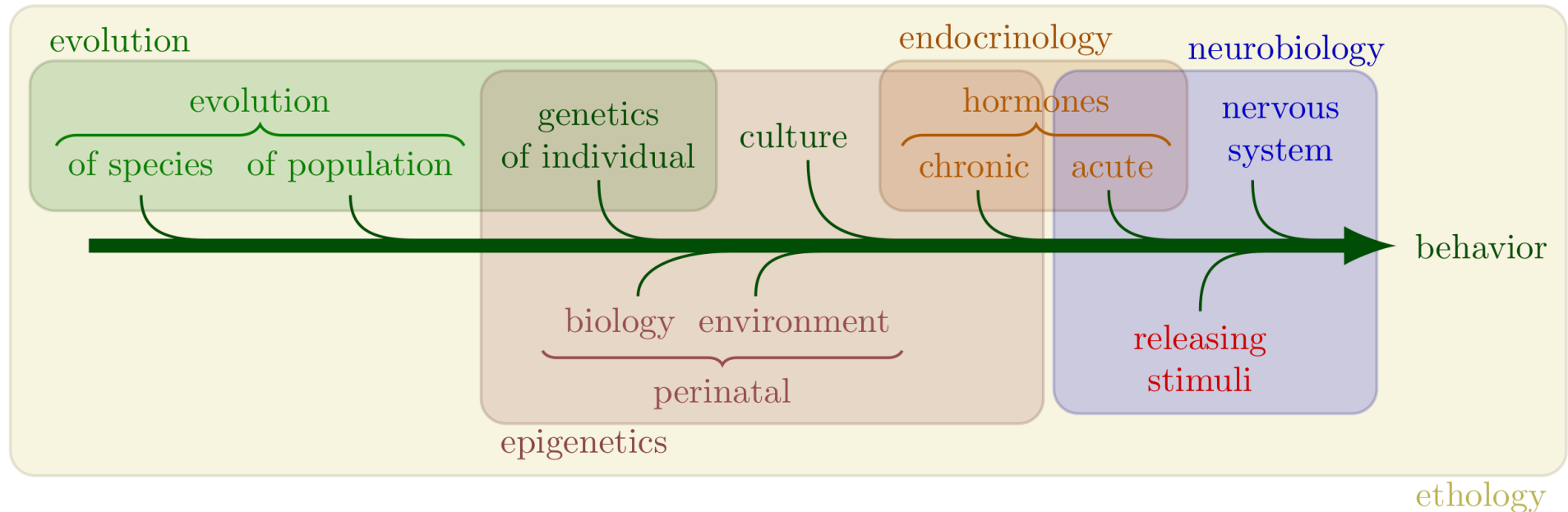


Hours to days before: hormones and bodily state

Hormones and physiological conditions alter how responsive the brain is. Relevant influences include:

- Stress; hunger; fatigue; pain; sexual arousal; illness; sleep deprivation.

Timeline of causes of behaviour

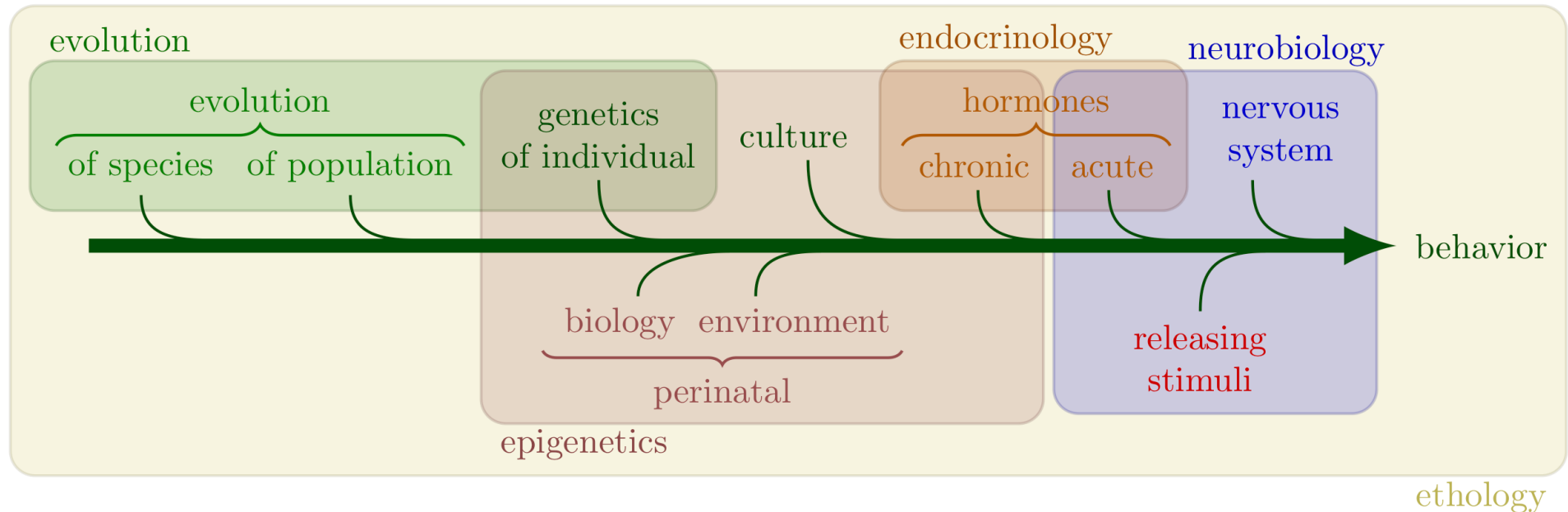


Days to months before: learning and neural plasticity

Recent experiences modify the nervous system. These may include:

- Reinforcement and punishment; repeated stress; social exclusion; successful cooperation; exposure to particular norms; practice controlling an emotional response.

Timeline of causes of behaviour

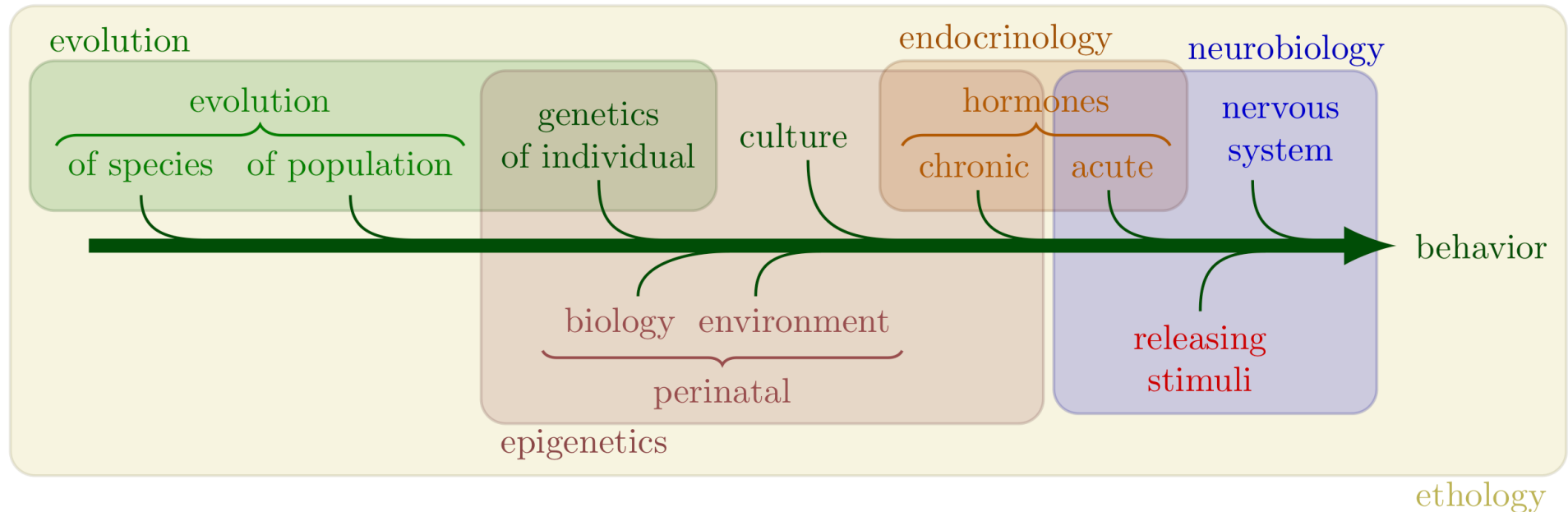


Years earlier: development

Childhood and adolescence shape the architecture and responsiveness of biological systems. Development affects:

- Stress regulation; attachment; emotional control; expectations about other people; sensitivity to threat; habitual ways of resolving conflict.

Timeline of causes of behaviour

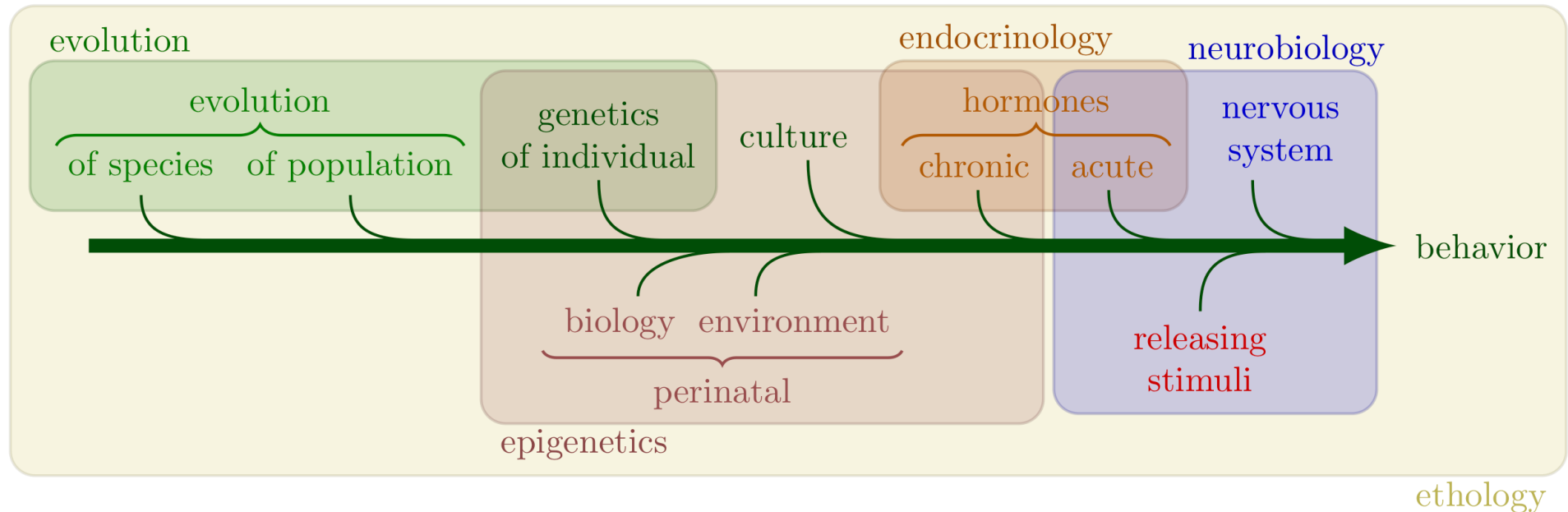


Generations earlier: culture

People inherit more than genes. They enter societies with established:

- Moral rules; gender expectations; hierarchies; categories of insiders and outsiders; systems of punishment and cooperation; stories about what behaviour is normal.

Timeline of causes of behaviour



Thousands to millions of years earlier: evolution

Natural selection shaped the capacities from which present behaviour emerges. Humans evolved capacities for:

- Aggression and competition; fear and threat detection; attachment and parental care; empathy and cooperation; punishment and reconciliation; group loyalty and suspicion of outsiders; learning, inhibition and behavioural flexibility.

A more nuanced view of the role of evolution in behaviour

Evolutionary history helped produce a capacity or sensitivity that develops and operates in interaction with present environments.

For example - aggression:

1. Evolution produces a flexible threat-detection system.
2. Culture teaches which people and actions count as threatening.
3. Childhood experience calibrates sensitivity to threat.
4. Recent stress increases physiological responsiveness.
5. An ambiguous comment is interpreted as hostile.
6. Neural activity produces an aggressive response.

Behaviour is biological, but biology is historical, developmental, contextual and interactive

Biology versus environment is a false opposition

- Environmental events alter hormones, gene expression, brain organisation and learning. Biology is partly the process through which environments affect behaviour.

Biological does not mean genetic

- A behaviour can have an immediate biological mechanism without individual differences in it being inherited.

Genetic does not mean inevitable

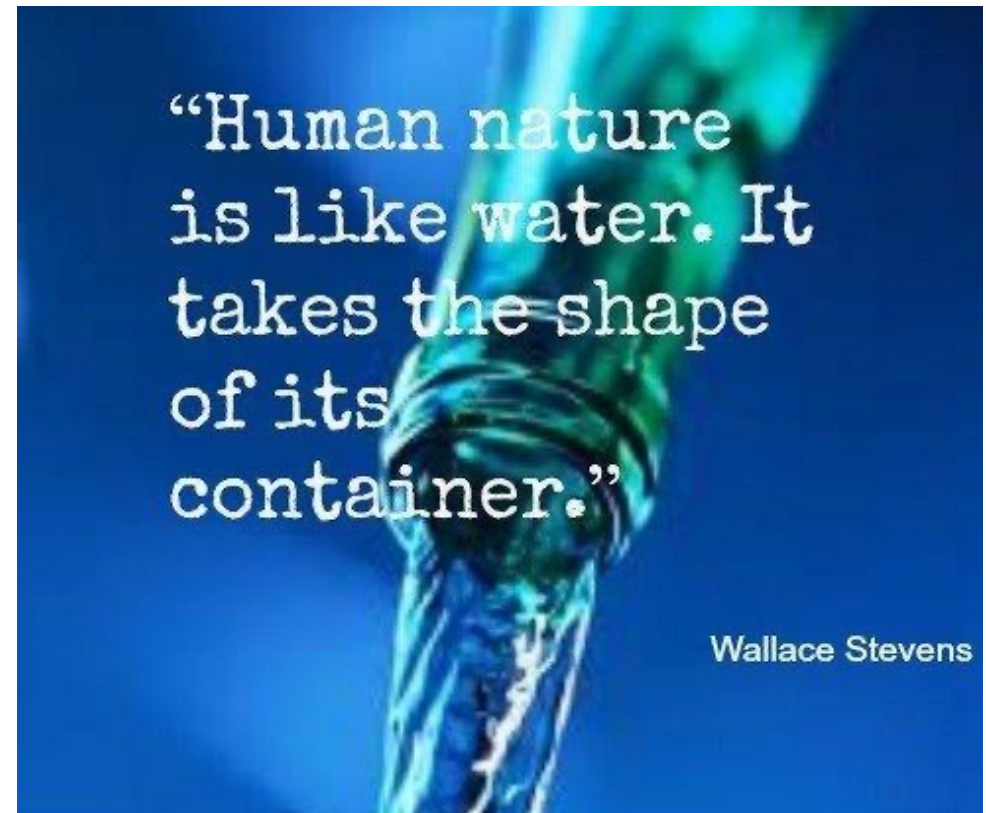
- Genes exert effects through development and particular environments. Genetic influences are usually probabilistic and conditional.

Evolved does not mean fixed

- Behavioural flexibility can itself be an evolved characteristic.

Explanation does not mean justification

- Understanding why violence or prejudice occurs does not make it morally acceptable. Explanation may instead help societies design more effective prevention and intervention.



Further readings

- **Maternal epigenetics**
 - Champagne F. A. (2018). Beyond the maternal epigenetic legacy. *Nature neuroscience*, 21(6), 773–774. <https://doi.org/10.1038/s41593-018-0157-6>
- **Gene-environment interactions:**
 - Dick DM. Gene-environment interaction in psychological traits and disorders. *Annu Rev Clin Psychol.* 2011;7:383-409.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC3647367/>
- **Sapolsky, R. M. (2017). Behave: The biology of humans at our best and worst. Penguin Books**
 - See video recording of a lecture: <https://www.youtube.com/watch?v=GRYcSuyLiJk>